Lightweight Machine Learning Models for Pneumonia Detection in Chest Radiographs

***Abstract*—This paper presents a lightweight machine learning (ML) framework for pneumonia detection from chest X-ray images. Three classical classifiers are evaluated: Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest (RF). The models are trained using 5,856 pediatric chest X-rays. Principal Component Analysis (PCA) is applied to reduce feature dimensionality. Among the evaluated models, the SVM achieves the best overall performance. It reaches 94.97% accuracy, an Area Under the Receiver Operating Characteristic Curve (AUC–ROC) of 0.9846, an Area Under the Precision–Recall Curve (AUC–PR) of 0.9939, and a Matthews Correlation Coefficient (MCC) of 0.8709. The model shows high sensitivity (97.54%) while maintaining improved specificity (88.01%). Class imbalance is explicitly handled during training. Robustness is evaluated using cross-validation, bootstrapping, and statistical significance testing. The framework produces calibrated confidence scores and supports visual interpretation of predictions. Due to low computational cost and stable behavior, the proposed approach is suitable for pneumonia screening in resource-limited clinical settings.**

***Index Terms*—Pneumonia Detection, Chest X-Ray, Machine Learning, Medical Imaging, Clinical Decision Support**

I. Introduction

Pneumonia remains a major cause of morbidity and mortality worldwide. In 2021, approximately 2.1 million pneumonia-related deaths occurred [1]. Children under five years of age and adults over seventy are the most affected populations. Mortality is highest in regions with limited access to trained radiologists and diagnostic infrastructure.

Chest X-ray imaging is the most widely used modality for diagnosing pneumonia. Interpretation, however, is subjective and time-consuming. Diagnostic outcomes depend on the reader's experience and workload. Delayed diagnosis and inter-observer variability remain common, particularly in under-resourced healthcare systems. As such, there is a great need for automated tools to support clinicians during routine screening.

A. Research Motivation and Problem

Pneumonia is one of the leading causes of death from lower respiratory tract infections. Mortality linked to antimicrobial resistance is driven in large part by *Streptococcus pneumoniae* [1]. The disease is not rare and is treatable. Yet outcomes remain poor in many settings, mainly because diagnosis is delayed or inconsistent.

Advances in AI, particularly deep learning (DL), have made automated medical image analysis possible. These diagnostic models often achieve high accuracy. Their resource demands are also high; large datasets, specialized hardware, and complex deployment pipelines are common requirements. These constraints limit use in low-resource clinical settings. There is a need for diagnostic models that are accurate, computationally efficient, and easy to deploy on standard clinical hardware.

B. Objectives and Contributions

This work presents a computationally efficient ML pipeline for automated pneumonia detection from chest X-ray images. Three interpretable classifiers are implemented and compared: SVM, KNN, and RF. The contributions of this work are threefold. First, it shows that a carefully tuned SVM can achieve strong diagnostic performance while using limited computational resources. Second, it presents a complete and reproducible pipeline that includes preprocessing, feature extraction, dimensionality reduction, and evaluation. Third, it reports calibration and clinical metrics that are directly relevant to screening scenarios.

C. Paper Organization

Section II reviews prior work on pneumonia detection. Section III describes the dataset, preprocessing steps, feature extraction, dimensionality reduction, and classification models. Evaluation metrics and the validation protocol are presented in Section IV. Section V discusses the experimental results with conclusions drawn in Section VI.

II. Related Work

Research on pneumonia detection from chest radiographs has followed several parallel directions. Some work prioritizes raw classification accuracy using DL models. Other studies combine deep feature extraction with lighter classifiers. A third direction focuses on classical ML methods to reduce computational cost and simplify deployment. This section reviews these directions and highlights their trade-offs.

A. DL Methods

Convolutional neural network (CNN)-based DL models dominate recent studies. These models operate directly on pixel data. Feature learning is implicit. In [2], transfer learning with VGG16, VGG19, ResNet50, and Inception-v3 was evaluated on pediatric chest X-rays, reaching 92.31% validation accuracy. Larger EfficientNet variants were examined in [3], where reported accuracy increased to roughly 94%.

Performance, however, is not consistent across datasets. When custom CNNs and ResNet-based transfer learning were applied to the RSNA pneumonia dataset, the reported peak accuracy dropped to 79% [4]. This gap suggests sensitivity to dataset characteristics, labeling protocols, or both.

Several approaches move beyond direct image classification. In one case, lung regions are segmented before classification using a transformer-based model, and accuracy exceeds 93% across multiple datasets [5]. Other work focuses less on architecture and more on the data. Class imbalance is handled using Synthetic Minority Over-sampling Technique (SMOTE) with pre-trained CNNs in [6], where VGG16 reaches 93.75% accuracy.

Interpretability appears in a smaller subset of studies. In [7], deep models are analyzed using multiple explainability techniques, including Layer-wise Relevance Propagation, with accuracy reported at approximately 91%.

Overall, DL models perform well. The cost is complexity. Large models, long training times, and hardware dependence remain common assumptions.

B. Hybrid Models Combining Deep and Classical Learning

To reduce end-to-end model complexity, several studies separate feature extraction from classification. In [8], lightweight CNN backbones such as ShuffleNet, NASNet-Mobile, and EfficientNet-b0 were used to extract features from 18,200 chest X-rays. Transfer learning alone produced an accuracy of 88%. Feature fusion was then applied, followed by classical classifiers including SVM and Extreme Gradient Boosting (XGBoost). This combination increased reported accuracy to 94%.

Localization is sometimes added to improve clinical interpretability. A hybrid pipeline combining CNN-based classification with You Only Look Once (YOLO) was proposed in [9]. The reported validation accuracy was 83%, with F1-scores of 0.799 for normal cases and 0.819 for pneumonia. Localization performance was reported separately, with an F1-score of 0.54 for pneumonia regions.

Hybrid pipelines reduce training cost compared to fully end-to-end deep models. Inference, however, still depends on deep feature extractors. This can remain a bottleneck in low-resource settings.

C. Classical ML Methods

Classical ML methods remain relevant where computational resources are limited. An early example is reported in [10], where an SVM was trained on chest X-ray images using PCA for dimensionality reduction. The reported accuracy was 71% on a standard CPU. While lower than deep learning results, the computational footprint was minimal.

Other classical approaches appear in different imaging modalities. In [11], logistic regression and linear regression were compared for pneumonia detection from MRI data. Logistic regression achieved higher accuracy, reaching 94% in that study.

Beyond image-based diagnosis, ventilator-associated pneumonia prediction was studied using the MIMIC-III database in [12]. An XGBoost model achieved an AUC–ROC of 0.94, and feature importance was analyzed using SHapley Additive exPlanations (SHAP).

Across prior studies, model choice is often influenced by deployment constraints rather than accuracy alone. DL approaches typically emphasize performance, whereas classical ML methods are selected to reduce computational and operational overhead.

III. Methodology

This section outlines how the pneumonia detection system is constructed and evaluated. The process begins with basic image preprocessing and continues through feature extraction and selection, dimensionality reduction, and classification. Three classical classifiers are considered in this study: SVM, KNN, and RF. Model size is not the primary concern. The focus is on accuracy, stability, and ease of deployment under limited computational resources. The complete workflow is shown in Fig. 1.

A. Methodological Framework

Each step is implemented independently and runs on standard computing hardware. The overall structure is designed to keep data handling and model training transparent.

a) Data ingestion and preprocessing.: A standardized pediatric chest X-ray dataset is acquired and checked for basic quality issues (e.g., unreadable scans or corrupted files). Labels provided by expert radiologists are preserved. All images are resized to 100×100 pixels, converted to grayscale, and intensity-normalized to reduce variation across scanners and acquisition conditions.

b) Feature extraction.: Features are computed from each image using multiple families of descriptors. Statistical moments summarize intensity distributions. Gradient-based features are extracted using Sobel operators. Texture descriptors are extracted using Local Binary Patterns. These features are selected because they are simple, interpretable, and computationally efficient.

c) Feature selection and dimensionality reduction.: Two lightweight selection steps are applied before dimensionality reduction. Variance thresholding removes near-constant features. Mutual information ranking keeps features that carry signal about the class label. PCA is then applied to compress the feature space.

d) Model training and hyperparameter search.: Three classifiers are trained: SVM, KNN, and RF. Training uses class-aware settings to mitigate the pneumonia–normal imbalance. Hyperparameters are tuned using 3-fold stratified cross-validated grid search.

e) Evaluation and statistical validation.: Model performance is characterized using Accuracy, AUC–ROC, AUC–PR, F1-score, and MCC. To examine robustness, additional resampling procedures are applied. These include 5-fold cross-validation, bootstrapping, and McNemar's tests for pairwise model comparison.

An overview of the full workflow, from image input to evaluation, is shown in Fig. 1.

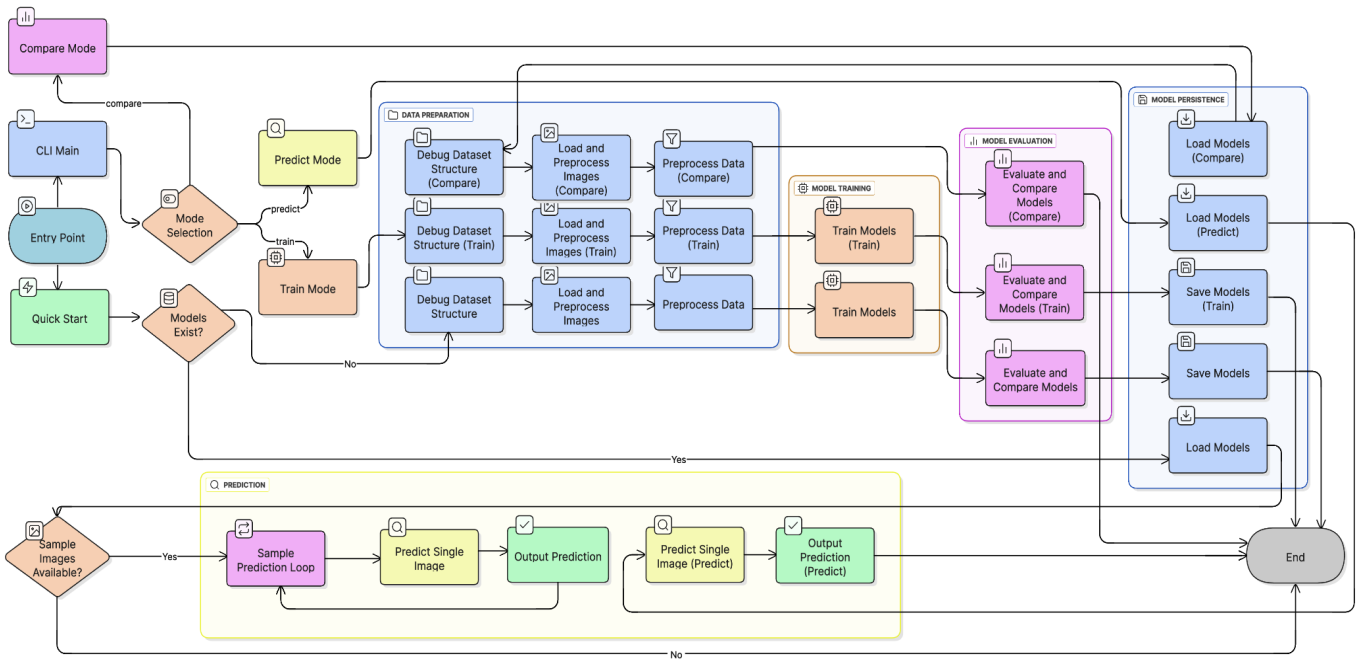

Fig. 1: Comprehensive workflow of the pneumonia detection system illustrating the complete pipeline from data acquisition through preprocessing, feature extraction, model training, evaluation, and deployment.

B. Dataset Description

1) Data Collection and Sources: The dataset comprises pediatric chest X-rays from the publicly available Pediatric Chest X-Ray dataset on Kaggle. Dataset images are anterior–posterior radiographs collected at Guangzhou Women and Children's Medical Center. Patients are children aged 1–5 years. Expert radiologists annotated each image as **NORMAL** or **PNEUMONIA** (bacterial or viral). Standardized splits are used: 5,216 training images, 624 test images, and 16 validation images.

Each image is processed using:

$$\Phi(I_{\text{raw}}) = \text{resize}\left(\text{normalize}(I_{\text{raw}}), (100, 100)\right) \tag{1}$$

where Φ maps raw images to a normalized grayscale representation used by the feature pipeline.

2) Data Statistics and Distribution: The dataset is imbalanced, which matches typical clinical prevalence. Out of 5,856 images, 4,273 (73.0%) are pneumonia and 1,583 (27.0%) are normal. The imbalance ratio is:

$$R = \frac{N_{\text{pneumonia}}}{N_{\text{normal}}} = \frac{4273}{1583} \approx 2.70 \tag{2}$$

The training set contains 3,875 pneumonia and 1,341 normal images. The test set contains 390 pneumonia and 234 normal images. The validation set is small (16 images) and is used for hyperparameter tuning. This class skew can bias a model toward the majority class, so training and evaluation are performed with imbalance-aware settings.

C. Image Preprocessing and Feature Engineering

Images are resized to 100×100 pixels using bicubic interpolation. They are converted to grayscale using:

$$I_{\text{gray}} = 0.2989R + 0.5870G + 0.1140B$$

Intensity normalization is performed using z-score normalization:

$$I_{\text{norm}} = \frac{I - \mu}{\sigma}$$

This step reduces acquisition-level variation.

Data augmentation is applied to the training set to improve robustness. Augmentation includes rotations ($\pm 15°$), flipping, and brightness adjustments. This expands the training set by a factor of eight.

Feature extraction uses a mix of simple descriptors. First-order statistics capture global intensity behavior. Sobel operators capture gradient structure. Local Binary Patterns capture texture. Pneumonia-specific measures are also computed to reflect coarse structural changes often visible in radiographs. Feature selection is performed using variance thresholding and mutual information ranking. PCA then reduces the feature space from approximately 30,000 dimensions to 50 components while preserving 81.27% of the variance. The resulting representation is standardized before training to maintain consistency across models.

D. Machine Learning Models

SVM, KNN, and RF are trained on the same PCA-reduced feature set (50 components). The models were selected because they are fast to train and infer, and easier to interpret than deep models. Training uses class-aware settings to mitigate

the 73/27 pneumonia–normal imbalance. Hyperparameters are tuned using cross-validation grid search to ensure a fair comparison.

1) SVM: The SVM classifier is implemented with a radial basis function (RBF) kernel. Hyperparameter selection is performed using stratified 3-fold cross-validation, which yields $C = 1.0$ and $\gamma = \texttt{scale}$. This setting avoids aggressive margin fitting while preserving non-linear decision boundaries.

Evaluation on the test set yields an accuracy of 94.97%. Sensitivity is high at 97.54%, and specificity reaches 88.01%. Threshold-independent measures show similar behavior, with AUC–ROC and AUC–PR values of 0.9846 and 0.9939, respectively. Probability outputs are comparatively stable, reflected in a log loss of 0.1315 and a calibration error of 0.0554.

Differences between training and test accuracy are limited to 2.17%. Across resampled folds, performance variation remains low. These results indicate that the SVM maintains a consistent balance between sensitivity, specificity, and probability calibration under the evaluated conditions.

2) KNN: The KNN classifier is evaluated using several neighborhood sizes and distance metrics. Hyperparameter tuning selects $n_neighbors = 11$ with Manhattan distance. This configuration favors local similarity while maintaining stable performance.

KNN achieves a test accuracy of 92.58%. Sensitivity is high at 97.43%, but specificity drops to 79.50%, leading to a higher false positive rate than SVM. MCC is 0.8073, reflecting weaker balance between classes.

Discriminative ability remains strong, with an AUC–ROC of 0.9727. However, probability calibration is less reliable than SVM. These results indicate that KNN prioritizes sensitivity at the expense of specificity, which may be undesirable in screening workflows where false positives increase clinical burden.

3) RF: The RF model consists of 100 decision trees. Hyperparameters are set to `n_estimators=100`, with no depth constraint and minimal split restrictions. This configuration maximizes model flexibility.

The model achieves a test accuracy of 92.06%. Sensitivity is high at 97.31%, but specificity is the lowest among the evaluated models at 77.92%. Training accuracy reaches 100%, resulting in a test gap of 7.94%, which indicates clear overfitting.

Despite this, discriminative performance remains high, with an AUC–ROC of 0.9744 and an AUC–PR of 0.9900. Calibration error is higher (0.1262) than both SVM and KNN. While RF handles non-linear feature interactions well, its overfitting behavior and reduced specificity limit its suitability for balanced clinical deployment.

4) Hyperparameter Optimization: Hyperparameters for all models are tuned using `GridSearchCV` with 3-fold stratified cross-validation. The objective is to maximize test accuracy while maintaining stable generalization.

For SVM, the search space includes $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\texttt{scale}, \texttt{auto}, 0.001, 0.01, 0.1\}$. The RBF kernel consistently outperforms linear alternatives. For KNN, neighborhood sizes between 3 and 11 are evaluated alongside different distance metrics. RF tuning explores tree counts between 50 and 200, as well as depth and split constraints. All models are trained and evaluated on the same PCA-reduced feature set. This ensures that performance differences reflect classifier behavior rather than feature representation.

IV. Experimental Setup

The evaluation protocol used to compare the three classifiers is outlined below. The goal is to assess classification performance, calibration behavior, and statistical reliability under identical test conditions.

A. Evaluation Metrics

1) Performance and Calibration Evaluation: Model performance is evaluated using a set of standard classification metrics. Accuracy measures overall correctness. Precision and recall capture class-specific behavior, with recall corresponding to sensitivity in the clinical context. Specificity reflects the ability to correctly identify normal cases. The F1-score summarizes precision and recall into a single value.

Threshold-independent discrimination is evaluated using AUC–ROC and AUC–PR. These metrics capture model behavior across different decision thresholds and are less sensitive to a fixed operating point. Class balance effects are further summarized using MCC, which accounts for all entries of the confusion matrix. Cohen's κ is also reported to measure agreement beyond chance.

Model calibration is evaluated to assess the reliability of predicted probabilities. Calibration curves are used to compare predicted confidence against observed outcomes. Expected Calibration Error (ECE) provides a quantitative summary of this alignment. Lower ECE values indicate better probability calibration.

Among the evaluated models, SVM exhibits the lowest calibration error (ECE = 0.055). KNN shows moderate miscalibration (ECE = 0.097), while RF displays the largest deviation (ECE = 0.126), particularly in mid-probability ranges. These differences are consistent with the performance trends reported in Table I.

B. Validation Protocol

All reported metrics are computed on the same held-out test set consisting of 624 images. Additional validation procedures are applied to assess stability and reduce variance.

Hyperparameter tuning relies on stratified 3-fold cross-validation. This choice preserves class proportions while keeping the search cost manageable. Once hyperparameters are fixed, 5-fold cross-validation is applied to probe sensitivity to data partitioning and quantify performance variability.

Uncertainty in the reported metrics is examined using bootstrapping. Pairwise differences between classifiers are then evaluated using McNemar's test to determine whether changes in prediction outcomes are statistically significant.

TABLE I: Comprehensive Performance Analysis of Pneumonia Detection Models

Model	Performance			Statistical			Clinical			Efficiency	
	Acc (%)	AUC–ROC	F1	MCC	Kappa	Log L	Spec (%)	PPV (%)	ECE	Train (s)	Inf (ms)
SVM	94.97	0.9846	0.9492	0.8709	0.8702	0.1315	88.01	95.64	0.0554	38.0	0.8
KNN	92.58	0.9727	0.9240	0.8073	0.8035	0.3120	79.50	92.76	0.0966	1.6	3.2
RF	92.06	0.9744	0.9185	0.7936	0.7891	0.2500	77.92	92.24	0.1262	62.0	1.5

Abbreviations: SVM = Support Vector Machine, KNN = K-Nearest Neighbors, RF = Random Forest, Acc = Accuracy, AUC–ROC = Area Under the ROC Curve, F1 = F1-Score, MCC = Matthews Correlation Coefficient, Spec = Specificity, PPV = Positive Predictive Value, ECE = Expected Calibration Error, Train = Training Time, Inf = Inference Time.
Note: All metrics are computed on the identical test set (624 images). Training times include full model fitting. Inference time is reported per image.

V. Results and Discussion

Experimental results are summarized in this section, followed by a discussion of observed performance differences between classifiers. Emphasis is placed on diagnostic reliability, error characteristics, and clinical relevance rather than raw accuracy alone.

A. Quantitative Results Analysis

Table I reports classifier performance on a common held-out test set. Accuracy varies within a narrow range. SVM records the highest value at 94.97%, followed by KNN (92.58%) and RF (92.06%). The differences are small, but they persist across multiple metrics.

All models show high sensitivity, exceeding 97%. Separation occurs on the negative class. SVM reaches a specificity of 88.01%, while KNN and RF remain below 80% (79.50% and 77.92%, respectively). This shift directly affects the number of false positive pneumonia predictions.

Threshold-independent metrics show a similar pattern. SVM attains an AUC–ROC of 0.9846 and an AUC–PR of 0.9939. KNN and RF produce slightly lower values. The same ordering appears in MCC, with scores of 0.8709 for SVM, 0.8073 for KNN, and 0.7936 for RF.

Differences also appear in probability behavior. SVM produces the lowest log loss (0.1315) and the smallest calibration error (0.0554). Calibration error increases for KNN and RF, indicating less stable probability estimates, particularly near the classification boundary.

B. Statistical Performance Comparison

Additional analysis is used to examine whether the observed differences persist under resampling. Accuracy confidence intervals are estimated using bootstrapping with 50 resamples. For SVM, the 95% interval spans [0.9533, 0.9712]. Intervals for KNN and RF overlap partially with this range but are centered at lower values.

Pairwise comparisons are evaluated using McNemar's test. Differences between SVM and the other models are statistically significant ($p < 0.05$). Variability across data splits is further examined through cross-validation. In this setting, SVM exhibits the lowest standard deviation across folds (0.52%), indicating more consistent behavior under resampling.

From a clinical perspective, high sensitivity (>97.3%) is maintained across all models. Specificity varies more strongly. The higher specificity of SVM (88.01%) reduces false positive predictions relative to KNN (79.50%) and RF (77.92%). This behavior is reflected in Positive Predictive Value (PPV), which reaches 95.64% for SVM compared to 92.76% for KNN and 92.24% for RF.

VI. Conclusion

The study evaluated the practicality of classical ML models (SVM, KNN, RF) for pneumonia detection using pediatric chest X-ray images. All models were trained and evaluated under identical conditions. Performance differences were driven mainly by specificity and probability calibration rather than sensitivity. In this setting, SVM reduced false positive predictions and produced more stable confidence estimates than the other models. These properties are important in screening scenarios. The findings indicate that classical approaches remain viable when computational resources are constrained. Future work will examine generalization to external datasets and real clinical deployments.

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